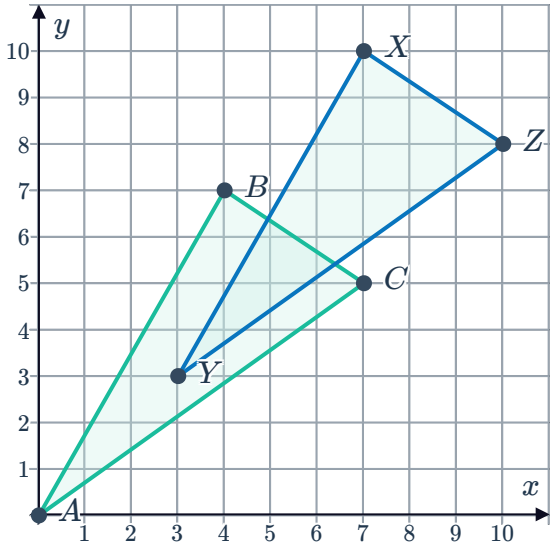
Coordinate methods and triangle congruence

1 Consider the following diagram.

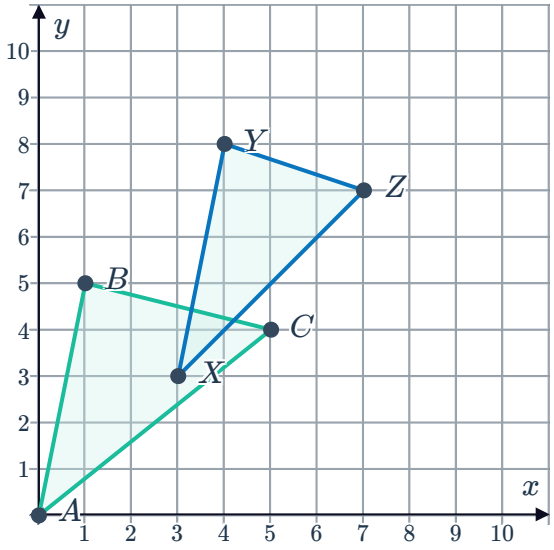


a Describe the translations that must be applied to move X to B.

b Hence, which points of $\triangle ABC$ will Y and Z translate to?

c Are $\triangle ABC$ and $\triangle YXZ$ congruent?

2 Consider the following diagram.



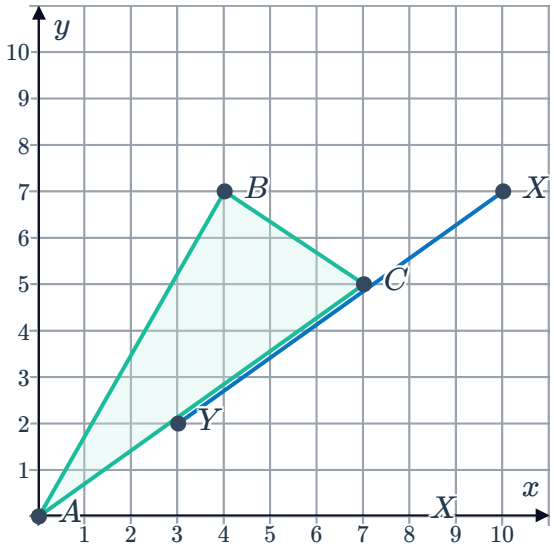
a Describe the translations that must be applied to move X to A.

b Hence, which points of $\triangle ABC$ will Y and Z translate to?

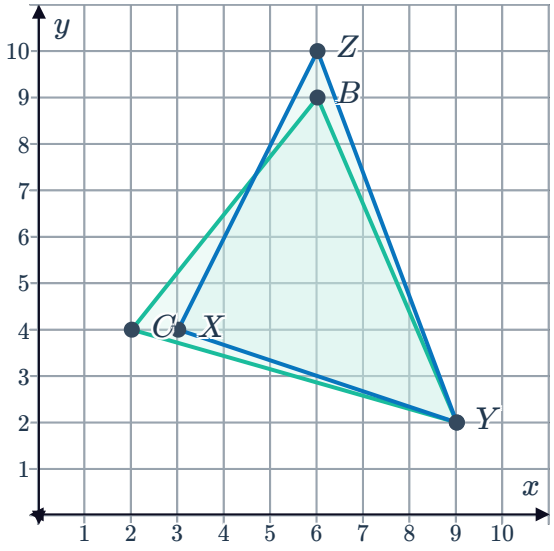
c Are $\triangle ABC$ and $\triangle XYZ$ congruent?

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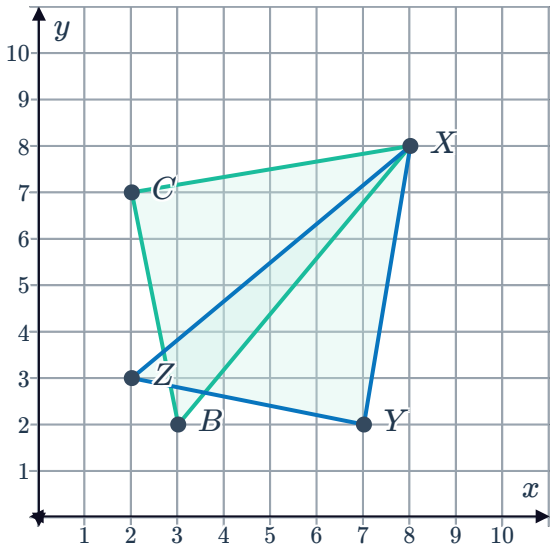
3 In the diagram shown, X is a translation of C, and Y is a translation of A.
Plot Z on the given coordinate plane so that the resulting $\triangle YZX$ is congruent to $\triangle ABC$.



4 Consider the two triangles drawn in the following diagram.
Are $\triangle YBC$ and $\triangle YXZ$ congruent?

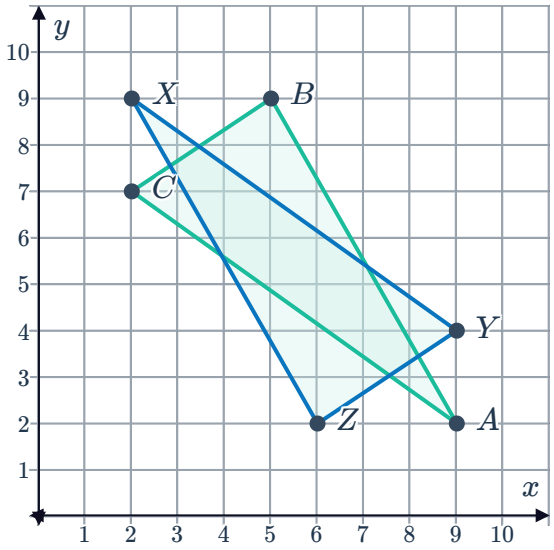


5 Consider the two triangles drawn in the diagram shown.
Are $\triangle XBC$ and $\triangle XYZ$ congruent? Explain your answer.



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6 Consider the two triangles drawn in the diagram below.
Are $\triangle ABC$ and $\triangle XZY$ congruent? Explain your answer.



7 For each of the following diagrams:

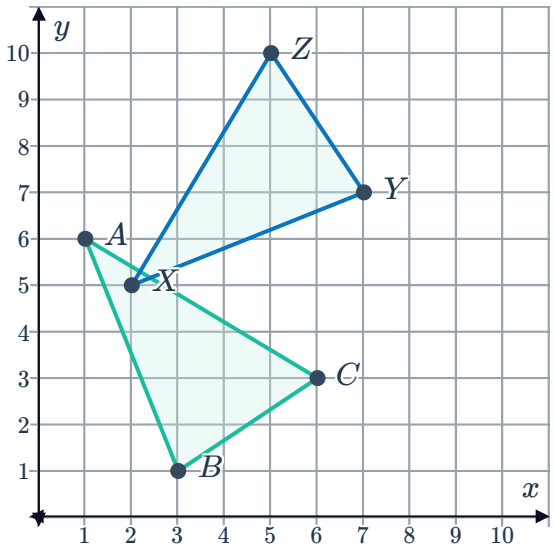
i Find AB.

ii Using the same method as the previous part, or otherwise, complete the following table of segment lengths.

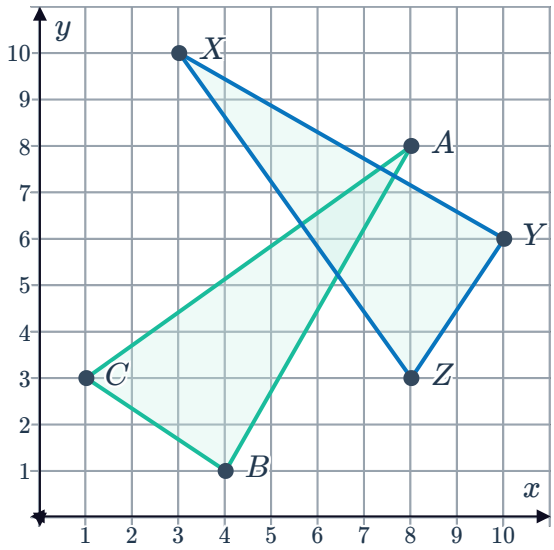
Segment	$\overline{AB}$	$\overline{BC}$	$\overline{CA}$	$\overline{XY}$	$\overline{YZ}$	$\overline{ZX}$
Length						

iii Are $\triangle ABC$ and $\triangle XYZ$ congruent?

a

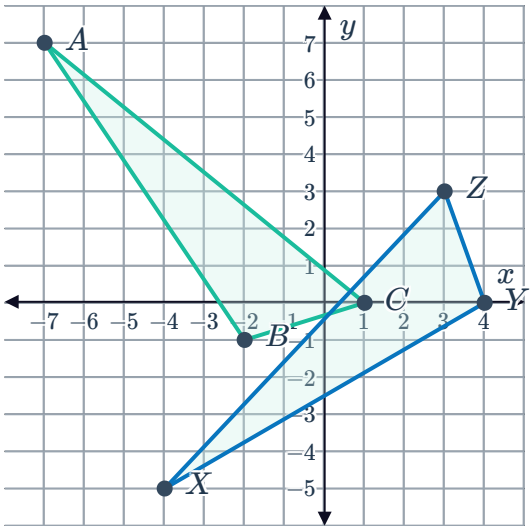


b

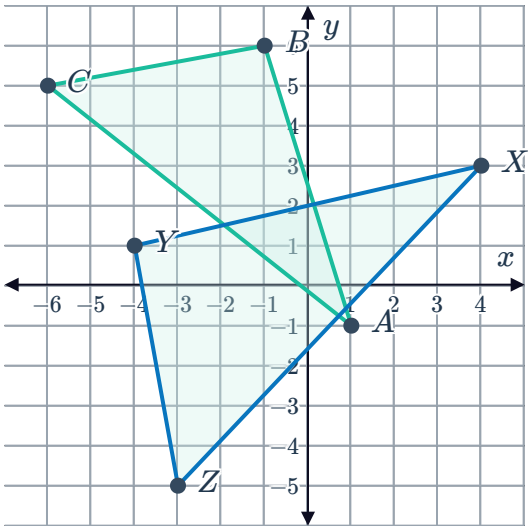


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c



d



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